

Investing in intelligence: AI human capital, wage inequality, and firm value in tourism, travel and hospitality

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ARTICLE INFO

Keywords:

Artificial intelligence
Human capital investment
Wage inequality
Firm value
Industry environment

ABSTRACT

Grounded in resource-based view and tournament theory, this study examines how AI human capital (AIHC) is associated with wage inequality and firm value in tourism, travel, and hospitality (TTH). Using an unbalanced panel of 1206 UK listed TTH firms comprising 7539 firm year observations from 2018 to 2024, we estimate internal instrument two step system GMM models with Windmeijer corrected standard errors. Findings indicate that AIHC is positively associated with firm value, negatively associated with wage inequality, and that wage inequality is negatively associated with firm value. Mediation analyses suggest that wage inequality partially channels the AIHC and firm value association. Moderation analyses show stronger AIHC and firm value associations in more dynamic and complex industries, whereas munificence shows no systematic moderating effect. Sectoral analyses reveal stronger value associations in travel and stronger inequality compression associations in hospitality. The study highlights workforce centered pathways for AI enabled competitive advantage.

1. Introduction

Tourism, travel, and hospitality (TTH) firms are increasingly deploying artificial intelligence (AI) to improve forecasting, pricing, capacity planning, personalization, and front-line service support. These applications are particularly consequential in service settings where value is co-produced with customers and performance depends on reliable coordination across people, processes, and information at scale (Buhalis & Law, 2008; Gretzel et al., 2015; Ivanov & Webster, 2019a, 2019b; Tussyadiah, 2020; Zhao et al., 2025). As a result, competitive advantage is increasingly shaped not only by access to AI tools, but also by the organisational capability to integrate AI into day-to-day service operations.

Despite the visibility of AI deployments in practice, tourism and hospitality scholarship still offers limited large-sample, firm-level evidence on whether AI capability translates into market-based value creation. Much prior work has been conceptual, case-based, or focused on traveller perceptions and adoption rather than on organisational outcomes such as firm value (Morosan & Dursun-Cengizci, 2024; Wang & Zhang, 2025). Moreover, empirical studies frequently proxy AI adoption using technology presence, capital expenditures, patents, or self reported adoption, which may be only imperfectly related to the

capabilities actually embedded in production and service delivery processes (Sevcenko et al., 2022; Verhoogen, 2023). This limits inference about how AI capability building affects performance in TTH and obscures potential distributional consequences inside firms.

To address this gap, we conceptualise AI capability as a human-capital resource and focus on AI human capital intensity, defined as firms' allocation of resources to attract, develop, and retain employees with AI-relevant skills and to embed these skills in complementary processes and governance (Liu & Zhang, 2025). Recent labour-market research operationalises AIHC using matched job-posting and resume data to identify AI-related skills and compute the share of AI-skilled workers at the firm-year level (Babina et al., 2023, 2024). In contrast to binary adoption indicators, this workforce-based measure captures both the intensity of AI capability and its diffusion within the organisation, which is central in labour-intensive service industries.

We develop a focused argument linking AIHC to both value creation and distributional outcomes. Drawing on the resource-based view (RBV), we expect AI-related skills to enhance firm value when they are integrated with complementary assets and work design, enabling better sensing, decision-making, and service innovation (Barney, 1991). However, building AI capability through the workforce may also reshape pay structures. The broader technology and labour literature

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<https://doi.org/10.1016/j.tourman.2026.105470>

Received 2 September 2025; Received in revised form 25 May 2026; Accepted 27 May 2026

Available online 11 June 2026

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offers competing predictions: skill-biased and task-based perspectives suggest that new technologies can increase the relative demand for high-skill labour and widen wage dispersion, whereas diffusion of skills and complementary upgrading can spread productivity gains more broadly (Acemoglu & Restrepo, 2019; Shen, 2022). Tournament theory further highlights that steep vertical pay dispersion can undermine morale, trust, and collaboration, which are critical to service quality and learning (Connelly et al., 2016; Lazear & Rosen, 1981; Ridge et al., 2015). Combining these perspectives, we examine whether AIHC is associated with lower intra-firm wage inequality, whether wage inequality is negatively associated with firm value, and whether inequality acts as a mechanism linking AIHC to value creation.

Because the returns to capability building are rarely uniform, we also examine boundary conditions that are salient in TTH. Industry environments differ in dynamism, complexity, and munificence, and these conditions shape both the value of AI-enabled sensing and coordination and the organisational challenges of deploying AI-related skills (Dess & Beard, 1984; Harrington & Kendall, 2005). We additionally examine whether the AIHC effects differ across hospitality, travel, and tourism subsectors, which vary in contact intensity, operational standardisation, and data availability. Accordingly, this study follows a concise problem-gap-contribution structure. It addresses a practical problem that is central to TTH: firms seek AI-enabled productivity and service improvements, yet AI investments may have meaningful distributional implications in a sector where work is labour intensive. It fills a research gap by providing large sample, longitudinal evidence on how workforce-based AI capability is associated with both firm value and wage inequality. It contributes to theory by integrating RBV and tournament theory to articulate a human-capital pathway for AI value creation and to identify contextual conditions that shape that pathway.

To address these gaps, we examine whether building AI capability through the workforce is associated with both firm performance and pay structures in tourism, travel, and hospitality. We compile a firm year dataset of 1206 UK listed TTH firms covering 2018 to 2024 (7539 observations), and we focus on the UK because listed firms disclose remuneration and financial information in a comparatively standardized way, making it a particularly informative setting for studying both wage inequality and market value (Department for Business, Energy & Industrial Strategy, 2018; UK Endorsement Board, 2021). We capture AI human capital intensity using a straightforward measure: the percentage of a firm's workforce with AI-related roles/skills in the prior year, which strengthens temporal ordering between AI talent investment and subsequent outcomes. We assess wage inequality using a widely adopted vertical pay measure, defined as the ratio of average top management compensation to estimated average employee pay. Firm value is measured using Tobin's Q, which is a standard market-based indicator. We also examine whether these relationships vary across industry conditions, including dynamism and complexity. By focusing on who holds AI skills and how broadly these skills are embedded within the organisation, our approach moves beyond binary notions of "AI adoption" and offers evidence on whether AI talent investment is associated with value creation and more equitable pay structures in labour-intensive service industries.

The remainder of the paper is organised as follows. Section 2 reviews the literature and develops the theoretical background. Section 3 presents the conceptual framework and hypotheses. Section 4 outlines the research design and econometric strategy. Section 5 reports the empirical results, and Section 6 discusses implications, limitations, and avenues for future research.

2. Literature review

This section synthesizes prior research that informs the study's research questions. We first review work on AI human capital intensity as a firm-level capability, with particular attention to how it is conceptualized, measured, and expected to affect value creation in

service settings. We then review evidence on wage inequality and pay dispersion, focusing on mechanisms through which technology-related skill investments may reshape within-firm pay structures. Across both subsections, we highlight points of convergence in the literature, clarify unresolved issues, and delineate how these gaps motivate the empirical questions examined in this study.

2.1. AI human capital

AI human capital intensity refers to the resources firms devote to attracting, developing, and retaining employees with AI-relevant competencies, such as machine learning, data science, and human-AI orchestration, and to embedding these competencies in complementary processes and governance (Liu & Zhang, 2025). Recent large-sample studies operationalize this construct by using matched job-posting and resume data to identify AI-related skills and to compute the share of AI workers at the firm-year level (Babina et al., 2023, 2024). This measurement approach moves attention from whether a firm uses AI in a general sense to the extent to which AI capabilities are embodied in the workforce. Theory suggests that the performance implications of AI human capital intensity depend on how AI-related skills complement the task structure of production. In services, AI capabilities can be mapped to different task intelligences, including mechanical, analytical, intuitive, and empathetic dimensions, implying that the returns to AI skills vary with the nature of service work and the degree of human interaction (Huang & Rust, 2018). A task-based perspective further predicts that AI investments can reallocate task content by automating some activities while creating new, higher-skill tasks, thereby changing workforce composition and returns to skills (Acemoglu & Restrepo, 2019). However, documented productivity paradoxes caution that benefits are not automatic. Value creation often depends on complementary assets, such as data availability, digital architecture, and work design, and may materialize only after diffusion and implementation lags (Brynjolfsson et al., 2017; Innocenti & Golin, 2022).

Within tourism, travel, and hospitality, discussions of AI capability have evolved from eTourism to smart tourism ecosystems, emphasizing digitally enabled coordination across firms, technologies, and customers (Buhalis & Law, 2008; Gretzel et al., 2015; Zhang et al., 2026). TTH is characterized by high contact intensity, volatile demand, and operational standardization, conditions under which AI-skilled employees can support both frontstage and backstage applications, including service robots and conversational agents as well as demand forecasting, pricing, and personalization (Ivanov & Webster, 2019a, 2019b; Li et al., 2025). Existing evidence also indicates that the objectives guiding automation, such as cost reduction versus service enhancement, shape customer responses and highlight the importance of aligning technical capability with service design (Belanche et al., 2021; Stylos et al., 2025). Adoption outcomes further depend on the interaction context, with acceptance shaped by perceived interaction quality, creepiness versus coolness, and perceived health risks during crises (Kim et al., 2021; Tussyadiah & Park, 2018).

To reflect current scholarly conversations in Tourism Management, we further position the study alongside recent work on generative AI, AI supported job crafting, AI chatbots, and AI based service failure recovery in tourism and hospitality (Li et al., 2025; Stylos et al., 2025; Wang et al., 2025; Zhang et al., 2024; Zhao et al., 2025). These studies clarify that AI now affects both customer facing service experiences and employee work design, whereas our study extends this stream by examining firm year variation in AI human capital, wage inequality, and market value.

Despite this progress, much of the TTH literature remains focused on technology presence and customer perceptions rather than treating AI human capital as a firm-level resource observable in workforce composition (Li et al., 2024). This limits inference about whether and how differences in AI skill shares are associated with firm value and with internal distributional outcomes. By employing labor-based, firm-year measures of AI human capital, the present study extends prior work by

linking talent investments to competitive outcomes and to pay structures within firms (Zhang et al., 2024). These arguments motivate our first research question on whether AI human capital intensity is associated with value creation in TTH firms, and they also set up the need to consider contextual boundary conditions examined later in the manuscript.

2.2. Wage inequality

Wage inequality, understood as differences in pay within and between firms, has re-emerged as a central feature of contemporary labor markets (Marsland, 2015). Two bodies of evidence are particularly relevant for firm-level research. First, matched employer-employee studies show that rising wage dispersion reflects both within-firm gaps and, to a substantial extent, widening differences between firms in the average wages they pay (Card et al., 2013; Song et al., 2019). Second, organizational research links pay dispersion to performance through competing mechanisms. Tournament theory suggests that larger pay spreads can motivate effort when promotion prospects and prize differentials are credible (Lazear & Rosen, 1981), while other work highlights potential costs through reduced collaboration, perceived unfairness, and higher turnover (Bloom, 1999; Shaw et al., 2002). Consequently, the relationship between inequality and performance is context dependent and likely to vary with task interdependence and skill complementarities (Guidi & Guardiancich, 2025).

In the UK, pay-ratio reporting underscores the prevalence of steep vertical pay structures. For 2023/24, the median chief executive earned 52 times the median employee salary across FTSE 350 firms (High Pay Centre, 2025). While pay ratios do not capture all dimensions of inequality, they signal differences between top management compensation and typical employee pay that may affect engagement, organizational cohesion, and governance concerns. These considerations are salient for research on AI-related skill investments because AI-driven work redesign can shift relative bargaining positions and the demand for complementary roles, potentially influencing within-firm wage structures.

These issues are especially pronounced in TTH, where employment is labor intensive, standardized, and demand volatile (Xu et al., 2025). Industry statistics show a persistent concentration of low pay in hospitality. In 2024, 53.6 percent of jobs in accommodation and food services were paid below the UK real Living Wage, representing the highest industry share for the thirteenth consecutive year (Living Wage Foundation, 2025). By comparison, 8.9 percent of all UK employee jobs were classified as low paid in 2023 using the OECD definition (ONS, 2023). Although pay growth accelerated in 2024 and accommodation and food services recorded the fastest year-on-year increase in median weekly earnings (9.8 percent), the sector remains characterized by low pay and variable hours (ONS, 2024). Vertical inequality is also evident at the top of the distribution, with pay-ratio disclosures indicating wide spreads in some travel-and-leisure firms, such as a CEO-to-median-employee ratio of 303:1 for Compass Group in 2023/24 (High Pay Centre, 2025). Research further documents gender pay differentials in hospitality, with estimates for European hospitality management roles ranging from 5.1 percent to 23.8 percent and links these gaps to occupational segregation and precarious work (de Castro Romero, Barroso, & Santero-Sánchez, 2025; Kronenberg & Fuchs, 2021).

Taken together, the literature indicates that TTH combines structural conditions that sustain unequal pay outcomes with ongoing technological and organizational change. However, existing work rarely connects firm-level investments in AI-related human capital to distributional outcomes within firms. This gap motivates our second research question on whether AI human capital intensity is associated with wage inequality in TTH firms, complementing the performance-focused question reviewed above.

2.3. Overarching theory

Resource-based view (RBV) (Barney, 1991, 1996) and tournament theory (Bognanno, 2001; Lazear, 1989) provide complementary lenses for explaining how AI human capital investment, wage inequality, and firm value are linked. RBV holds that heterogeneous, difficult-to-imitate resources and capabilities to orchestrate them underpin advantage (Barney, 1991, 1996; Peteraf, 1993; Teece et al., 1997; Wernerfelt, 1984). Human capital, particularly specialized and firm embedded expertise, is a central intangible resource that can meet VRIN conditions when combined with idiosyncratic routines and path dependence (Lepak & Snell, 1999; Ployhart & Moliterno, 2011). In AI-intensive contexts, the relevant resource is not merely the presence of algorithms but the stock and renewal of AI talent that designs, integrates, and scales models with data, architecture, and domain knowledge (Li et al., 2024). Under RBV, increasing the share of AI-skilled employees can raise value creation via innovation, efficiency, and analytics-driven decisions, elevating Tobin's Q (Bin-Nashwan & Li, 2025). Yet RBV also stresses appropriation hazards: when value creation depends on mobile experts, rents may be dissipated through bargaining and turnover unless governance and pay systems support knowledge integration and reduce hold-up risks (Coff, 1997; Fosso Wamba et al., 2024; Sirmon et al., 2007; Teece, 2007).

Tournament theory models compensation as prizes that elicit effort in rank-order contests (Lazear & Rosen, 1981; Rosen, 1986). Larger spreads between organizational tiers can motivate contestants when output is noisy and monitoring is difficult. However, research on pay dispersion shows that, beyond modest spreads, inequality can impair cooperation, undermine fairness perceptions, and raise coordination and turnover costs effects that are salient where tasks are interdependent (Bloom, 1999; Cowherd & Levine, 1992; Pfeffer & Langton, 1993; Shaw et al., 2002). Accumulated evidence suggests that performance consequences of dispersion hinge on context (Connelly et al., 2014; Wade et al., 2006). For top management teams (TMTs), extreme vertical pay gaps damage collaboration with the broader workforce, offsetting incentives.

Integrating these theories clarifies our study's relationships. First, RBV predicts that building a larger, denser stock of AI talent creates a rarer resource base and stronger capabilities for sensing, seizing, and transforming opportunities in data-rich, dynamic markets (Teece, 2007). Second, as firms broaden AI expertise beyond the apex, value creation becomes more distributed across technical and cross-functional roles (Hughes et al., 2025). Organizations that depend on collaborative AI pipelines (i.e., data engineering, MLOps, product, operations) recalibrate pay to recruit and retain scarce mid-tier specialists, narrowing TMT-to-employee ratios (Yuan et al., 2025). RBV's appropriation logic implies that diffusing specialized knowledge can reduce bargaining-power asymmetries, attenuating extreme prize spreads. Third, tournament theory implies potential incentive benefits, but in AI-enabled service organizations, where innovation and service quality hinge on knowledge sharing, excessive dispersion depresses cooperation and learning, harming value capture (Bamberger et al., 2021). AI human capital therefore enhances value directly (via capabilities) and indirectly by reshaping rewards in ways that improve collaboration and reduce coordination frictions (Liu & Zhang, 2025). Sirmon et al. (2007) emphasize that value realization requires resource orchestration, structuring, bundling, and leveraging human assets. Pay systems play a critical role. When wage dispersion aligns with levels conducive to interdependence, firms are better able to translate AI knowledge into scalable routines, thereby enhancing Tobin's Q. In turbulent environments, the returns to AI talent and collaborative capacity are amplified, thereby strengthening the AI-value relationship in line with the resource-based view and dynamic capabilities perspective.

3. Conceptual framework and hypotheses development

Guided by the resource-based view and tournament theory (Barney, 1991, 1996; Bognanno, 2001; Lazear, 1989), we developed a model that links firms' AI human capital intensity to wage inequality and firm value. RBV motivates the core pathway: AI human capital constitutes a firm-specific, hard-to-imitate resource that enhances innovation, process efficiency, and decision quality, thereby generating an expected positive association with firm value. Tournament theory clarifies the incentive and coordination consequences of the pay structure: excessive vertical pay dispersion undermines collaboration and knowledge sharing, depressing value creation. Integrating these lenses, our model posits (i) a direct positive association between AI human capital and firm value; (ii) a negative association between AI human capital and wage inequality; and (iii) a negative association between wage inequality and firm value, yielding partial mediation of the AI-value link through inequality. We further contend that environmental dynamism, complexity, and munificence shape the returns to AI human capital.

3.1. AI human capital intensity and firm value

Investments in artificial intelligence human capital, reflected in an expansion of the share of employees with AI-relevant skills, constitute a strategic resource with the potential to enhance firm value when effectively embedded in complementary organisational assets (Liu & Zhang, 2025). Drawing explicitly on the resource-based view, AI-related expertise in areas such as data engineering, machine learning, and human-AI orchestration can satisfy the conditions of value, rarity, imperfect imitability, and non-substitutability when combined with firm-specific data, architectures, and domain knowledge (Barney, 1991; Peteraf, 1993). In this sense, AI human capital represents a firm-level capability rather than a standalone technology input.

Dynamic capabilities theory further clarifies how AI human capital is translated into value creation. By enhancing firms' abilities to sense opportunities, seize them through analytics-enabled decision making, and transform service and operational processes, AI skilled employees may help organizations convert technical potential into competitive advantages that are reflected in market valuations such as Tobin's Q (Teece, 2007; Teece et al., 1997). Recent measurement advances identify AI workers through matched job postings and worker profiles, allowing the construction of firm-year shares of AI talent that have been shown to predict innovation and growth outcomes (Babina et al., 2024). The primary value-creation channels are both product- and process-based, as AI human capital is expected to support predictive analytics, personalization, and revenue management while also streamlining operations, thereby aligning with higher expected cash flows and lower uncertainty (Shollo et al., 2022).

However, the resource-based view also emphasizes that returns to AI human capital are conditional rather than automatic. First, value realization depends on complementarities with organisational assets and often unfolds with diffusion and implementation lags, consistent with the productivity paradox associated with advanced digital technologies (Brynjolfsson et al., 2017). Second, appropriation challenges arise because AI-skilled employees are relatively mobile and their contributions are difficult to fully specify ex ante. Without effective resource orchestration and incentive alignment, value may dissipate through turnover or bargaining dynamics (Coff, 1997; Sirmon et al., 2007). Third, misalignment between AI capabilities and work design can undermine coordination, collaboration, and organisational learning, thereby attenuating performance gains (Adler & Borys, 1996; Argote & Miron-Spektor, 2011).

These mechanisms are particularly salient in tourism, travel, and hospitality, where demand volatility, high contact intensity, and operational interdependence amplify the importance of analytics-enabled decision making and service design. AI human capital with expertise in data engineering, machine learning, and human-AI orchestration

supports revenue management, personalization, and service operations under such conditions, thereby enhancing firm value when embedded in complementary routines and data infrastructures (Ivanov & Webster, 2019a, 2019b; Mariani & Borghi, 2021; Tussyadiah, 2020). Empirical evidence in hospitality and tourism shows that AI adoption improves forecasting accuracy, pricing strategies, and guest experiences, but that performance gains are contingent on complementary capabilities and organisational redesign (Buhalis & Law, 2008; Gretzel et al., 2015; Huang & Rust, 2018; Ivanov & Webster, 2019a, 2019b; Li et al., 2025; Stylos et al., 2025; Zhang et al., 2026). Consistent with a human-capital-based explanation, recent labour-based measures indicate that higher firm-level shares of AI-skilled workers are associated with innovation and growth outcomes (Babina et al., 2024). Taken together, these arguments suggest that AI human capital intensity operates as a capability-building mechanism that is expected to be associated with firms' ability to create value in data-intensive and service-oriented contexts. Accordingly, we propose.

H1. AI human capital intensity is positively associated with firm value.

3.2. AI human capital intensity and wage inequality

Investments in artificial intelligence human capital, reflected in a growing share of employees with expertise in data, machine learning, and human-AI orchestration, can reshape firms' internal pay structures (Brynjolfsson & McElheran, 2019; Mo et al., 2024; Tambe et al., 2020). This study's distributional predictions are grounded in tournament theory, which emphasizes how vertical pay differentials affect coordination, incentives, and performance within firms.

Tournament theory suggests that while steep pay spreads may motivate effort in settings with independent tasks, they can undermine cooperation, learning, and retention when production is highly interdependent (Bloom, 1999; Cowherd & Levine, 1992; Pfeffer & Langton, 1993; Shaw et al., 2002). These coordination costs are especially salient in AI-intensive service contexts, where effective deployment relies on tightly coupled pipelines spanning data engineering, model development, operations, and frontline service delivery (Huang & Rust, 2018). As firms expand their AI-skilled workforce, the bargaining power of mid-tier technical and cross-functional specialists increases, heightening the need for compensation systems that support collaboration and limit appropriation hazards (Coff, 1997; Li et al., 2019).

Empirical evidence using labour-based measures of AI skills shows that higher firm-level shares of AI-skilled employees are associated with changes in workforce composition that support innovation and growth (Babina et al., 2024). From a tournament-theoretic perspective, maintaining excessive vertical pay dispersion in such contexts weakens coordination and value capture, whereas more compressed pay structures facilitate effective orchestration of AI-related capabilities (Acemoglu & Restrepo, 2022; Bloom, 1999; Shaw et al., 2002; Wu et al., 2024). Accordingly, we propose.

H2. AI human capital intensity is negatively associated with wage inequality within firms.

3.3. Wage inequality and firm value

Wage inequality between top managers and employees has long been debated in relation to firm value. Tournament theory argues that wider pay spreads can motivate effort and ambition by strengthening advancement incentives, thereby enhancing performance when rewards are perceived as attainable (Bidwell et al., 2013; Lazear & Rosen, 1981; Rosen, 1986). However, organizational research highlights substantial costs associated with excessive pay dispersion. Large pay gaps can undermine perceptions of fairness, erode trust, and weaken collaboration, particularly in interdependent production settings (Cowherd & Levine, 1992; Pfeffer & Langton, 1993). Empirical evidence links high inequality to lower quality, reduced satisfaction, and higher turnover,

which in turn depress firm value (Bloom, 1999; Shaw et al., 2002). Within top management teams, extreme pay disparities further impair coordination and execution (Wade et al., 2006). Consistent with this evidence, meta-analyses suggest that while modest inequality may enhance incentives, its marginal benefits diminish and eventually reverse as dispersion increases (Bamberger et al., 2021; Connelly et al., 2014).

These trade-offs are particularly salient in tourism, travel, and hospitality, where value creation depends on service quality, coordination, and knowledge sharing across organizational levels. Excessive wage inequality can erode employee morale and cooperative climates that are essential for delivering integrated guest experiences (Cheng & Wong, 2015). In addition, high pay ratios expose firms to reputational and legitimacy risks, as stakeholders may perceive extreme executive compensation as misaligned with good governance, thereby weakening brand trust and market valuation (Bebchuk & Fried, 2004; Fiss et al., 2012). In such contexts, wage inequality functions both as an internal coordination challenge and as an external legitimacy liability that constrains firms' ability to translate resources into sustainable value creation. Accordingly, we propose.

H3. Wage inequality is negatively associated with firm value.

3.4. The mediating role of wage inequality

Wage inequality can operate as a transmission mechanism linking AI human capital intensity to firm value by shaping internal coordination and incentive alignment rather than by directly generating capabilities. As AI-related work is organised around interdependent roles spanning data engineering, machine learning, and operational deployment, effective collaboration becomes critical for converting technical inputs into organisational outcomes (Huang & Rust, 2018). In such settings, steep vertical pay differentials can impede integration by weakening cooperation and increasing retention risks among pivotal non-executive specialists. Firms that expand their AI-skilled workforce therefore face incentives to recalibrate compensation structures toward these roles, reducing excessive top-to-employee pay gaps in order to sustain coordination and limit appropriation hazards (Barney, 1991; Coff, 1997; Peteraf, 1993).

Tournament theory further clarifies why such adjustments matter for value creation. While modest pay dispersion can motivate effort, excessive inequality undermines perceptions of fairness, erodes trust, and weakens coordination, thereby increasing turnover and agency costs and depressing value capture (Bloom, 1999; Cowherd & Levine, 1992; Pfeffer & Langton, 1993; Shaw et al., 2002). Although a skill-biased technological change perspective predicts that scarce AI skills may widen wage gaps (Acemoglu & Restrepo, 2019), recent firm-level evidence shows that higher shares of AI workers are associated with workforce shifts that support innovation and growth, consistent with broader diffusion of value creation beyond the apex (Babina et al., 2024).

These mechanisms are especially salient in tourism, travel, and hospitality, where value creation relies on tightly coupled frontstage and backstage processes and where coordination costs of high pay dispersion are particularly pronounced. By compressing excessive wage inequality, firms can improve cooperation, knowledge sharing, and retention, thereby enhancing their ability to translate AI-related capabilities into superior market performance (Connelly et al., 2014; Wade et al., 2006). Taken together, these arguments suggest that AI human capital intensity reduces wage inequality, which in turn enhances firm value, implying a mediating relationship.

H4. Wage inequality mediates the relationship between AI human capital intensity and firm value.

3.5. The moderating role of industry environment

Industry environment conditions (i.e., dynamism, munificence, and complexity) are contingencies that shape how firms capture value from AI human capital investments (Mostafiz et al., 2025). Contingency theory and the strategic management literature argue that resource–performance relationships depend on external turbulence, slack, and information-processing demands (Dess & Beard, 1984; Lawrence & Lorsch, 1967). In dynamic environments, marked by unpredictable changes in demand and technology, the marginal returns to AI talent rise because analytics, experimentation, and adaptive routines enable superior sensing and seizing opportunities (Teece, 2007). Complexity likewise increases the payoff to AI human capital by raising the value of data integration and cross-functional coordination, so AI-rich firms can process more information and orchestrate responses more effectively (Keats & Hitt, 1988; Miller & Friesen, 1983). Munificence is theoretically more ambiguous. Resource abundance can ease complementary investments and capability reconfiguration, particularly in interdependent TTH service systems (Barney, 1991; Huang & Rust, 2018; Ivanov & Webster, 2019a, 2019b; Peteraf, 1993; Teece et al., 1997), yet slack can also weaken discipline, slow diffusion, and buffer coordination failures (Castrogiovanni, 1991).

Hospitality research similarly suggests that capability–performance links strengthen most clearly when uncertainty and coordination demands are high (Harrington, 2001a, 2001b; Jogaratnam & Tse, 2006; Tajeddini, Ratten, & Denisa, 2020). We therefore expect dynamism and complexity to amplify the AIHC–value relationship, while any positive role of munificence is more likely to emerge when resource richness facilitates complementary investment under sufficiently dynamic or complex conditions. This RBV–contingency view clarifies why similar AI investments yield heterogeneous value across TTH firms (Lalicic & Weismayer, 2021). AI-enabled forecasting and personalization become more valuable as uncertainty rises (Kumar et al., 2019). Therefore, we suggest that.

H5a. The positive relationship between AI human capital intensity and firm value is stronger when industry dynamism is higher.

H5b. The relationship between AI human capital intensity and firm value is expected to become more positive under higher industry munificence, particularly when munificence supports complementary investment under dynamic or complex conditions.

H5c. The positive relationship between AI human capital intensity and firm value is stronger when industry complexity is higher.

4. Methodology

The United Kingdom provides a particularly suitable context for examining the performance and distributional implications of investment in artificial intelligence human capital in tourism, travel, and hospitality. Tourism and hospitality are economically and socially significant in the UK, and official labour-market statistics show a large accommodation and food service workforce alongside persistent vacancies, underscoring both the scale of the sector and the salience of staffing frictions (Office for National Statistics, 2025a, 2025b).

Beyond sector scale, the UK setting is methodologically advantageous. Listed firms operate under strong governance and disclosure requirements, including standardized remuneration reporting and CEO-to-employee pay-ratio disclosure, which supports consistent measurement of executive compensation and firm outcomes over time (Department for Business, Energy & Industrial Strategy, 2018; UK Endorsement Board, 2021). Service delivery in tourism, travel, and hospitality is also labour intensive and knowledge interdependent, so labour costs represent a material determinant of firm performance. Industry benchmarks show that payroll per available room increased by 8.5 percent in the first quarter of 2025 relative to the same period in 2024, with

London hotels experiencing a 4.6 percent year-on-year increase in payroll per available room, underscoring the sensitivity of operating margins to staffing dynamics (HotStats, 2025; Knight Frank, 2025).

These institutional and labour-market characteristics make firm-year variation in AI staffing, measured as the share of AI-skilled workers, particularly informative for understanding value creation in the sector, as analytics-enabled work design and automation increasingly complement scarce human capital. Together, these features facilitate consistent benchmarks for top-management pay, labour expenses, and firm value across firms and over time. Although the empirical analysis focuses on UK firms, the underlying theoretical mechanisms are expected to generalise to other service economies, with the magnitude of effects varying according to institutional characteristics such as wage-setting systems, employment protection, and disclosure regimes.

4.1. Sample and data collection

We constructed an industry-focused panel of UK-listed tourism, travel, and hospitality firms for 2018–2024 to examine how AI human capital intensity relates to wage inequality and firm value. The initial sampling frame comprised FTSE 350 constituents classified to TTH-relevant sectors. To enhance longitudinal coverage and account for entries, exits, and index reconstitutions, we augmented the frame with firms listed on the London Stock Exchange that operate in closely related TTH activities (e.g., accommodation, restaurants, leisure travel, tour operators, online platforms, and ancillary services). We identified candidates using industry codes and business descriptions and verified

eligibility via annual reports. After matching and cleaning, the final dataset includes 1206 unique firms and 7539 firm-year observations, yielding an unbalanced panel consistent with firm turnover and missing disclosures over time.

We use the term UK listed TTH firms to describe the cumulative universe of firms with UK equity market listings or trading records during the observation window rather than a single current FTSE 350 cross section. The final count is therefore larger than a current index snapshot because it combines historical entrants, exits, AIM and Main Market listings, and multiple TTH adjacent activities after manual verification of business descriptions and annual reports. The broad coverage is analytically useful because the study examines an unbalanced firm year panel across seven annual waves, and Table 1 reports how the 1206 unique firms are distributed across accommodation, food and beverage, travel services, transportation, leisure and attractions, gaming, and ancillary services.

Table 1 shows the study profile. The panel spans the sector's principal subsectors (i.e., Accommodation, Food & Beverage, Travel Services/OTAs, Transportation, Leisure & Attractions, Gaming & Casinos, and Ancillary Services), providing broad coverage of business models and scale. Units of analysis are firm-years and industry classification follows 2-digit SIC, which also underpins construction of the environmental moderators (dynamism, munificence, complexity). Inclusion criteria require availability of top-management compensation, labour expenses, employee headcount, and market/financial data, and firm-years with missing essentials are excluded, as documented in the data-availability panel. Financials are reported in GBP, continuous variables are

Table 1
Sample profile and study characteristics.

Panel A. Overall Sample Profile								
Characteristic		Value				Notes		
Total firms (unique)		1206				Distinct firms across 2018–2024 Unbalanced panel; one observation per firm per year Seven annual waves		
Total firm-year observations		7539						
Observation window		2018–2024						
Panel B. Subsector composition								
Subsector	Example SIC (2-digit)	Number of Firms (N)	% of Firms	Firm-years (N)	% of Firm-years	Median Years/Firm	Median Employees	Median Annual Sales (GBP)
Accommodation (Hotels & Resorts)	55	280	23.2%	1781	23.6%	6	3000	£0.90bn
Food & Beverage (Restaurants/Pubs)	56	260	21.6%	1603	21.3%	6	10,000	£1.20bn
Travel Services & OTAs	79	190	15.8%	1156	15.3%	6	2000	£0.50bn
Transportation (Airlines/Airports)	45	110	9.1%	764	10.1%	7	20,000	£3.50bn
Leisure & Attractions	90	200	16.6%	1256	16.7%	6	5000	£0.80bn
Gaming & Casinos	92	86	7.1%	515	6.8%	5	12,000	£1.80bn
Ancillary Services (Payments/Tech)	62	80	6.6%	464	6.2%	5	1200	£0.35bn
Panel C. Coverage by Year								
Year	Firms Observed (N)				Firm-year Observations (N)			
2018	995				995			
2019	1050				1050			
2020	1048				1048			
2021	1080				1080			
2022	1110				1110			
2023	1128				1128			
2024	1128				1128			
Panel D. Data availability and exclusions								
Criterion	Firm-years (N)		% of Total (7,539)		Notes			
Missing TMT compensation	146		1.94%		Unavailable or incomplete remuneration disclosures No breakdown of staff costs Headcount not reported or inconsistent No reliable price/shares for the fiscal year Winsorized rather than deleted Reassigned or removed after manual review			
Missing labour expense	98		1.30%					
Missing employee headcount	121		1.60%					
Missing market data (price/shares)	84		1.11%					
Outliers truncated (winsorization)	0		0.00%					
Sector misclassification (eligibility)	43		0.57%					

winsorized at conventional tails, and AI human capital enters lagged ($t-1$) to strengthen temporal ordering. Year and industry fixed effects are applied to absorb period-specific shocks and time-invariant sectoral heterogeneity. Together, the sample profile indicates wide representation across TTH activities, sufficient within-firm time variation for dynamic estimation, and transparent quality screens, supporting comparability and reducing bias from missingness and extremes.

4.2. Measures

Our study employed several measures to operationalize the core constructs (i.e., AI human capital investment, wage inequality, firm value, and industry environment). These variables were carefully defined and computed to ensure construct validity, comparability across firms, and theoretical alignment. To ensure alignment between our empirical design and the research questions, we selected measures that directly map onto the study's core constructs. Our independent variable captures the extent to which firms build AI capability through workforce composition, which is central to examining whether AI human capital intensity relates to subsequent firm value and wage inequality outcomes. Our dependent variables capture, respectively, market-based value creation and within-firm vertical pay structure, which together reflect the study's focus on both performance and distributional implications.

AI Human Capital Investment. AI human capital was proxied by the share of AI-related workers in total employment at the firm-year level. This measure is appropriate because it captures the intensity of AI capability embedded in the workforce, which is the central mechanism through which the study theorises effects on value creation and wage structures. Following and Babina et al. (2024) and Liu and Zhang (2025), we operationalize AI human capital as the firm-year share of employees with AI-relevant roles or skills, which we interpret as AI human capital intensity in the workforce. We use this measure as a revealed indicator of firms' investment outcomes in recruiting, developing, and retaining AI-relevant talent, rather than as a direct accounting measure of AI spending. Importantly, this operationalization does not presume that AI is fully embedded in frontline service delivery across all tourism, travel, and hospitality firms. Even when firms rely on externally developed systems, effective deployment in AI-enabled services requires internal capabilities for data and integration work, analytics-enabled work design, governance, and human-AI orchestration, particularly given task interdependence in service settings (Huang & Rust, 2018). This perspective is consistent with the sector's evolution from eTourism toward smart tourism ecosystems, in which value creation increasingly depends on firm-level digital and analytics capabilities spanning frontstage and backstage processes (Buhalis & Law, 2008; Gretzel et al., 2015; Ivanov & Webster, 2019a, 2019b; Li et al., 2025; Mariani & Borghi, 2021; Stylos et al., 2025; Tussyadiah, 2020; Zhang et al., 2026).

To measure AI human capital intensity (AIHC), we identify AI-skilled workers by matching AI-related skill tags in worker profiles and job postings to individuals' employment records. AI-related skills are defined using a curated dictionary of AI and machine-learning competencies, including core methods (e.g., machine learning, deep learning, neural networks, reinforcement learning), application domains (e.g., natural language processing, computer vision, recommender systems), and common AI toolkits and deployment skills (e.g., TensorFlow, PyTorch, Keras, scikit-learn, and MLOps-related tools). Generic skills that are not specific to AI (e.g., spreadsheet skills or programming languages without an AI/ML context) are not sufficient for AI classification. Skill extraction is applied consistently across the full dataset using the same dictionary and rules, after which profiles and postings are linked to firms through standardized employer identifiers and aggregated to the firm-year level to compute AIHC as the percentage share of AI-skilled workers in total employment. To strengthen face validity, Appendix A supplements this procedure with role-title crosswalk and role-evolution

examples that translate the captured skill tags into concrete TTH job families and use cases. Appendix A, Table A4 presents the AI-related skill taxonomy, detailing the skill categories, representative keywords, and classification rules used to identify AI-skilled workers through a dictionary-based, rule-driven text analysis. Appendix A, Table A5 outlines the step-by-step deterministic linkage of classified worker profiles and job postings to firms and the aggregation procedure used to construct firm-year AI human capital investment, while Appendix A, Table A7–A8 clarify how those classified roles map into practice. We used the following equation to construct AI human capital investment:

$$\text{AI Human Capital Intensity}_{it} = \left(\frac{\text{Number of AI Skilled Employees}_{it}}{\text{Total Employees}_{it}} \right) \times 100 \quad (1)$$

Where, AI Human Capital Intensity_{it} represents the percentage share of employees with AI related skills within firm *i* in year *t*.

Because our panel begins in 2018, we provide additional descriptive evidence to assess whether the AI human capital measure captures meaningful variation in the early years of the study period. Following Liu and Zhang (2025) and Babina et al. (2024), our measure reflects AI-related workforce composition, capturing AI-relevant roles and skills rather than assuming widespread, fully internal AI applications in frontline service delivery from 2018 onward. This distinction is important in tourism, travel, and hospitality, where AI-related capabilities often reside in analytics and operational support functions that enable revenue management, forecasting, and personalization, and where AI value creation depends on integrating tools and routines across interdependent service processes (Buhalis & Law, 2008; Gretzel et al., 2015; Huang & Rust, 2018). Appendix A, Table A6 shows a clear upward trend in AI-related staffing and in the share of firm-years with non-zero AI employment across segments, while Appendix A, Table A7–A8 clarify that the early AIHC counts are concentrated in small analytics, data-engineering, and revenue-management teams that later broaden into recommender-system, conversational-AI, MLOps, and other deployment roles. Together, these descriptive checks indicate substantial and widening adoption of AI-related human capital across industries and over time.

Wage Inequality. The pay-ratio measure is relevant because it directly captures within-firm vertical pay structure between senior leadership and employees, aligning with the study's distributional focus and with policy-relevant disclosure practices for listed firms. Pay dispersion was measured as the ratio of the average top management team (TMT) wage to the average employee wage (Bamberger et al., 2021; Faleye et al., 2013). Average TMT compensation was calculated by dividing the total remuneration of top managers including fixed salaries and long-term incentives by the number of TMT members (Connelly et al., 2016; Ridge et al., 2015). The denominator was the average employee wage, computed as total labour expenses net of TMT compensation divided by total employees. We opted for the TMT rather than CEO pay, as the latter is highly volatile and less representative of broader managerial compensation structures. Our study used the following equation to construct wage inequality:

$$\text{Wage Inequality}_{it} = \frac{\text{Average TMT Compensation}_{it}}{\left(\frac{\text{Total Labour Expenses}_{it} - \text{Total TMT Compensation}_{it}}{\text{Number of Employees}_{it}} \right)} \quad (2)$$

Where:

$$\text{Average TMT Compensation}_{it} = \frac{\text{Total TMT Compensation}_{it}}{\text{Number of TMT Members}_{it}} \quad (3)$$

Firm Value. Tobin's Q is used because it reflects market expectations about future performance and intangible value, which is well suited for assessing whether investments in AI-related skills are associated with

value creation beyond short-run accounting outcomes. Firm value was assessed using Tobin's Q, an established indicator of long-term performance and market-based expectations of intangible resources (Lang & Stulz, 1994; Tobin, 1969). Given the complexity of exact computation, we adopted Chung and Pruitt's (1994) simplified approximation:

$$Q_{it} = \frac{\text{Market Value of Equity}_{it} + \text{Preferred Stock}_{it} + \text{Total Debt}_{it}}{\text{Total Assets}_{it}} \quad (4)$$

Where: Q_{it} , represents Tobin's Q for firm i in year t , which captures the market-based valuation of the firm relative to its asset base.

where MVE is the market value of equity, PS is preferred stock, DEBT is net debt, and TA is total assets. This approximation explains more than 96% of variation in Tobin's Q and has been widely used in strategic management research.

Industry Environment. To capture contextual moderators, we adopted widely used measures (Keats & Hitt, 1988; Palmer & Wiseman, 1999; Zhu et al., 2021). Dynamism was assessed as the volatility of industry sales over five years, obtained by regressing the log of industry sales on time and using the standard error of the slope coefficient. Munificence was proxied by industry sales growth over five years, based on the regression slope of logged sales against time. Complexity was measured using the reciprocal of the Herfindahl index, so that higher values indicate greater heterogeneity and lower concentration. These environmental dimensions allow us to contextualize how industry structure and turbulence condition the association between AI human capital and firm value. These moderators are included because the theorised relationships are expected to vary with environmental uncertainty, resource conditions, and institutional and competitive constraints that shape firms' ability and incentives to deploy AI-related human capital.

Control variables: to account for potential confounding influences, we incorporated a set of control variables at the firm, industry, and temporal levels. At the firm level, we included size, age, and profitability, as these characteristics are consistently identified in prior research as key determinants of both organizational performance and growth trajectories (Bardhan et al., 2013; Dezső & Ross, 2012; Julian & Ofori-Dankwa, 2017; Kortmann et al., 2014; Lam et al., 2016). Firm size was proxied by the natural logarithm of annual sales (Bardhan et al., 2013), while firm age was measured as the natural logarithm of the number of years since founding, calculated as the difference between the current year and the firm's establishment year (Dezső & Ross, 2012; Li et al., 2010). Profitability was captured using return on assets (ROA), reflecting earnings relative to total assets (Lam et al., 2016). In addition to firm-level characteristics, we included a set of year dummies to control for unobserved time-specific shocks, and industry dummies to account for sectoral heterogeneity in environmental conditions and competitive dynamics. Collectively, these controls help to isolate the effects of the focal independent variables on firm value by reducing bias from observable and unobservable firm and context-specific factors. Appendix A shows study variables conceptualisation and measurements.

4.3. Generalized method of moments (GMM) approach

Consistent with prior work on organizational outcomes (Lam et al., 2016; Mukherji et al., 2011; Zhu et al., 2021), we adopt a dynamic panel data (DPD) approach in which the one-period lag of the dependent variable enters the model as a regressor. This term captures path dependence (i.e., performance in $t-1$ influencing performance in t) and improves the fidelity of short-run adjustment dynamics. In addition, we align the timing of covariates and outcomes to preserve temporal ordering. Key predictors, including AI human capital investment, are measured at year t , whereas firm value (Tobin's Q) is observed at $t+1$. Accordingly, each specification includes both the lagged dependent variable and a one-year lead of the outcome relative to the predictors.

This design raises three econometric concerns. First, including the lagged dependent variable alongside firm fixed effects induces

correlation between regressors and the composite error term, producing the well-known dynamic panel bias (Nickell, 1981). Second, our panel is unbalanced because some firm-years lack inputs required to construct Tobin's Q or wage inequality, a common feature of disclosure-based datasets. Third, potential endogeneity affects key regressors (i.e., AI human capital may shape wage inequality and firm value) but feedback from organizational performance to strategic choices (e.g., investment in AI capabilities) is also plausible (Bardhan et al., 2013; Lam et al., 2016). Ignoring such reverse causality and simultaneity would risk overstating the impact of AI human capital on the outcomes of interest.

To address these issues, we developed the models using the system generalized method of moments (system GMM) estimator of Arellano and Bover (1995) and Blundell and Bond (1998). System GMM jointly exploits a differenced equation (instrumented with lagged levels) and a levels equation (instrumented with lagged differences), generating internal instruments that are orthogonal to the fixed effects and thus mitigating dynamic panel bias (Roodman, 2009). The estimator is well suited to unbalanced panels with endogenous regressors and has been characterized as one of the most robust technologies for such contexts (Flannery & Hankins, 2013). In our application, potentially endogenous variables (e.g., AI human capital, wage inequality) are treated as endogenous or predetermined and instrumented with their appropriate lags, while strictly exogenous controls enter uninstrumented. We emphasize that this is an internal-instrument strategy rather than a claim of a separately identified external-IV design for AIHC; because no exclusion restriction appeared sufficiently defensible in this setting, we avoid forcing a weak external instrument and instead triangulate the dynamic estimates with static firm fixed-effects benchmarks reported in Appendix C. We implement instrument-set management (e.g., collapsing the instrument matrix and limiting lag depth) to curb proliferation and safeguard the reliability of over-identification tests, following best practice (Roodman, 2009).

In the reported specifications, the GMM style instrument window for the lagged dependent variable, AIHC, WAGE, and interaction terms is restricted to lags t minus 2 and t minus 3 in the differenced equation. The corresponding levels equation uses lagged first differences as internal instruments, while year and industry indicators are entered as standard instruments. The instrument matrix is collapsed and lag depth is capped in all main models, yielding 120 to 130 instruments for the Tobin Q specifications and 115 instruments for the WAGE specification, which is far below the 1206 firm groups. Strictly exogenous controls are not expanded as GMM style instruments. Appendix B, Table B1 and Appendix C, Table C6 report the instrument counts, alternative lag caps, and proliferation diagnostics.

All estimations are conducted in Stata 16 using two-step system GMM with finite-sample, robust standard errors. The two-step estimator improves efficiency and is robust to heteroskedasticity; we apply the Windmeijer finite-sample correction to adjust the downward bias in two-step standard errors (Roodman, 2009). As diagnostic checks, we report the Arellano-Bond tests for serial correlation in the first-differenced residuals and the Hansen test of over-identifying restrictions and acceptable p-values on these statistics provide evidence consistent with valid instruments and correct model specification. Collectively, this estimation strategy aligns the identification approach with the theoretical timing of effects, while directly confronting dynamic bias, unbalanced panels, and endogeneity with an estimator designed for such settings.

4.4. Model specifications

In line with the research design, we estimated dynamic panel data models to examine the direct, mediating, and moderating relationships. Each specification includes a lagged dependent variable to capture persistence, lagged explanatory variables to strengthen temporal ordering, year fixed effects, and industry fixed effects.

Model 1. Direct effect of AI human capital intensity on firm value

$$Q_{it} = \alpha_0 + \alpha_1 Q_{i,t-1} + \alpha_2 AIHC_{i,t-1} + \alpha_3 SIZE_{i,t-1} + \alpha_4 AGE_{i,t-1} + \alpha_5 PRFT_{i,t-1} + \lambda_t + \eta_{j(i)} + \varepsilon_{it}. \quad (5)$$

Model 2. Mediating role of wage inequality

Because mediation requires both the mediator equation and the outcome equation, the mediation model is specified as follows.

$$WAGE_{it} = \beta_0 + \beta_1 WAGE_{i,t-1} + \beta_2 AIHC_{i,t-1} + \beta_3 SIZE_{i,t-1} + \beta_4 AGE_{i,t-1} + \beta_5 PRFT_{i,t-1} + \lambda_t + \eta_{j(i)} + u_{it}. \quad (6a)$$

$$Q_{it} = \theta_0 + \theta_1 Q_{i,t-1} + \theta_2 AIHC_{i,t-1} + \theta_3 WAGE_{i,t-1} + \theta_4 SIZE_{i,t-1} + \theta_5 AGE_{i,t-1} + \theta_6 PRFT_{i,t-1} + \lambda_t + \eta_{j(i)} + \nu_{it}. \quad (6b)$$

Model 3. Moderating role of the industry environment

$$Q_{it} = \varphi_0 + \varphi_1 Q_{i,t-1} + \varphi_2 AIHC_{i,t-1} + \varphi_3 ENV_{j(i),t-1} + \varphi_4 (AIHC_{i,t-1} \times ENV_{j(i),t-1}) + \varphi_5 SIZE_{i,t-1} + \varphi_6 AGE_{i,t-1} + \varphi_7 PRFT_{i,t-1} + \lambda_t + \eta_{j(i)} + \xi_{it}. \quad (7)$$

where Q_{it} denotes Tobin's Q for firm i in year t ; $AIHC_{i,t-1}$ denotes lagged AI human capital intensity; $WAGE_{i,t-1}$ denotes lagged wage inequality; $SIZE$, AGE , and $PRFT$ denote firm size, firm age, and profitability, respectively; $ENV_{j(i),t-1}$ denotes the relevant industry environment dimension, namely dynamism, munificence, or complexity, for the industry j to which firm i belongs; λ_t denotes year fixed effects; $\eta_{j(i)}$ denotes industry fixed effects; and ε_{it} , u_{it} , ν_{it} , and ξ_{it} are idiosyncratic error terms.

4.5. Endogeneity

Endogeneity is a first-order concern in our setting because the focal outcomes and strategic choices are jointly determined over time, the data are observational, and several regressors may be measured with error. Following established econometric guidance (Angrist & Pischke, 2009; Wooldridge, 2010), we consider three primary sources: (i) dynamic persistence (past firm value influencing current outcomes), (ii) omitted time-varying firm- and industry-level factors, and (iii) simultaneity and reverse causality, whereby firms adjust AI human capital investment and compensation structures in response to performance (Roberts & Whited, 2013; Wintoki et al., 2012). To address these concerns, we adopt an identification strategy that aligns econometric design with the theoretical timing of effects.

First, we estimate dynamic panel models that include the lagged dependent variable to capture persistence. Because this term is correlated with unobserved firm effects, standard pooled OLS and fixed-effects estimators are biased in short panels (Nickell, 1981). We therefore employ the system GMM estimator (Arellano & Bover, 1995; Blundell & Bond, 1998), which combines equations in differences and levels and uses internal lagged instruments. This approach mitigates dynamic panel bias while improving efficiency through additional moment conditions (Bond, 2002; Roodman, 2009).

Second, we treat AI human capital intensity and wage inequality as endogenous (or predetermined) and instrument them using deeper lags within the system GMM framework. Firm-level controls are treated as predetermined, while year and industry fixed effects absorb common shocks and time-invariant heterogeneity. Regressors are lagged to preserve temporal ordering, and internal instruments are constructed using deeper lags (e.g., $t-2$, $t-3$). Importantly, we do not rely on an external instrumental variable for AIHC. A credible external IV that satisfies both

relevance and exclusion restrictions is difficult to justify in this context, as most plausible instruments would directly affect firm value. Instead, identification relies on temporal ordering, internal instrumentation, and diagnostic validation, which are more defensible in this setting. Specifically, AIHC, WAGE, the lagged dependent variable, and interaction terms are instrumented using the collapsed lag window t minus 2 to t minus 3. Firm size, age, and profitability enter with one-year lags as controls, and year and industry effects enter as standard instruments. This specification is reported consistently in the notes to Table 3 and in Appendix B.

Third, to limit finite-sample bias and instrument proliferation, we follow best practices in system GMM implementation (Flannery & Hankins, 2013; Roodman, 2009). Specifically, we collapse the instrument matrix, restrict lag depth, avoid instrumenting strictly exogenous variables, and ensure that the number of instruments remains well below the number of firms. Estimation is conducted using two-step system GMM with Windmeijer-corrected standard errors.

We assess model validity using standard diagnostics, including Arellano-Bond tests for serial correlation and Hansen and difference-in-Hansen tests of over-identifying restrictions. Across specifications, AR (2) tests fail to reject the null of no second-order serial correlation, and Hansen tests yield acceptable p-values, supporting instrument validity. Additional robustness checks vary lag structures, instrument sets, and sample composition, supporting the view that results are not artifacts of outliers or specific time periods. Moreover, we complement the dynamic analysis with static panel estimations (pooled OLS, random effects, and firm fixed effects) reported in Appendix E (Table E1). These models yield directionally consistent results, although coefficients are more conservative in magnitude, reflecting the absence of dynamic adjustment and explicit treatment of endogeneity. The comparison highlights that part of the cross-sectional association reflects persistent firm heterogeneity, reinforcing the need for dynamic modeling. Additional supporting evidence is provided through placebo lead tests and alternative lag specifications (Appendix E, Table E2), which show that future AIHC does not predict current outcomes and that lagged effects remain stable but attenuated. Together, these results are consistent with a temporally ordered relationship rather than reverse causality. Taken together, while no observational design can fully eliminate endogeneity concerns, the combination of dynamic specification, internal instrumentation, diagnostic testing, and benchmark static models substantially strengthens identification. Accordingly, the reported relationships should be interpreted as robust evidence consistent with the theorized mechanisms, while acknowledging the usual limitations of observational firm-level research.

5. Results

5.1. Descriptive statistics

Table 2 reveals bivariate patterns that accord with our theorized mechanisms while remaining modest in magnitude, as expected for firm-year data. AI human capital (AIHC) is positively related to firm value (Tobin Q; $r = 0.12$, $p < 0.01$) and negatively related to wage inequality (WAGE; $r = -0.18$, $p < 0.01$), whereas WAGE is negatively associated with Tobin Q ($r = -0.20$, $p < 0.01$). Collectively, these correlations align with H1–H3 and are consistent with the proposed partial mediation (H4), although inferential interpretation rests on the dynamic panel results reported subsequently. Industry conditions show the anticipated signs: dynamism (DYN) correlates positively with Tobin Q (0.12, $p < 0.01$) and AIHC (0.10, $p < 0.01$) and weakly negatively with WAGE (-0.05 , $p < 0.05$); complexity (CMX) and munificence (MUN) also exhibit small positive correlations with Tobin Q (0.08, $p < 0.05$; 0.09, $p < 0.05$, respectively) and AIHC (0.06, $p < 0.05$; 0.08, $p < 0.05$). Among controls, profitability (PRFT) shows the strongest association with Tobin Q (0.30, $p < 0.01$), supporting its inclusion as a covariate, while SIZE and AGE display weaker positive relations with Tobin Q (0.07, $p < 0.05$;

Table 2
Correlation matrix.

Variables	Mean	Std. dev	Tobin Q	AIHC	WAGE	DYN	CMX	MUN	SIZE	AGe	PRFT
Tobin Q	0.72	2.68	1								
AIHC	0.08	0.27	0.120**	1							
WAGE	91.07	108.55	−0.200**	−0.180**	1						
DYN	1.70	0.80	0.120**	0.100**	−0.050*	1					
CMX	1.20	0.60	0.080*	0.060*	−0.030	0.220**	1				
MUN	1.60	0.70	0.090*	0.080*	−0.020	0.100**	0.080*	1			
SIZE	14.20	1.10	0.070*	0.020*	0.120**	0.060*	0.040	0.050*	1		
AGe	3.40	0.60	0.050*	0.060*	0.080	0.020	0.010	0.020	0.085	1	
PRFT	0.07	0.09	0.300**	0.100**	−0.100**	0.090*	0.060*	0.070*	0.120**	0.077	1

Table 3
Dynamic panel (system GMM) results: Direct, mediation, and moderation.

Variables	Model 1 (Tobin Q)	Model 2 (Tobin Q + WAGE)	Model 3 (Tobin Q × DYN)	Model 4 (Tobin Q × CMX)	Model 5 (Tobin Q × MUN)	Model 6 (WAGE)
Constant	0.720** (19.80)	0.818** (23.65)	0.819** (24.70)	0.818** (23.66)	0.807** (22.60)	—
Lagged DV	0.624*** (14.90)	0.601*** (14.30)	0.615*** (14.70)	0.613*** (14.60)	0.612*** (14.55)	0.702*** (18.06)
AIHC(t-1)	0.065** (2.50)	0.061* (1.84)	0.060* (1.71)	0.061* (1.92)	0.060* (1.98)	−0.064** (2.50)
WAGE (t-1)	—	−0.162** (2.25)	—	—	—	—
SIZE (t-1)	0.092 (1.60)	0.069 (1.28)	0.071 (1.27)	0.068 (1.29)	0.070 (1.22)	0.096 (1.29)
Age (t-1)	0.081 (2.01)	0.077 (1.05)	0.062 (1.89)	0.074 (1.82)	0.092 (1.78)	0.072 (1.88)
PRFT (t-1)	0.310*** (4.43)	0.298*** (4.19)	0.301*** (4.22)	0.299*** (4.20)	0.297*** (4.17)	−0.090** (2.36)
Dynamism (DYN) (t-1)	—	—	0.080** (2.11)	—	—	—
AIHC × DYN (t-1)	—	—	0.012** (2.40)	—	—	—
Complexity (CMX) (t-1)	—	—	—	0.016* (1.30)	—	—
AIHC × CMX (t-1)	—	—	—	0.023** (2.52)	—	—
Munificence (MUN) (t-1)	—	—	—	—	0.043 (1.20)	—
AIHC × MUN (t-1)	—	—	—	—	0.090 (1.17)	—
Years FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7539	7539	7539	7539	7539	7539
Instruments	120	125	130	130	130	115
AR(1) p-value	0.000	0.000	0.000	0.000	0.000	0.000
AR(2) p-value	0.214	0.231	0.247	0.251	0.258	0.188
Hansen J p-value	0.356	0.387	0.412	0.431	0.404	0.421
Diff-in-Hansen p-value	0.482	0.459	0.463	0.474	0.468	0.497

Notes.

- L. Dependent variable denotes the one-period lag of the dependent variable. ENV is a composite industry environment index (dynamism, munificence, complexity) standardized at the industry-year level.

- Dependent variable by column: Models 1–5 = Tobin Q; Model 6 = WAGE. Coefficients reported with t-statistics in parentheses. Year and industry effects included in all models.

- The two-step system GMM with Windmeijer-corrected standard errors; instruments collapsed, and lag depth limited. Model 5 reports the average linear munificence interaction; conditional follow-up analyses are reported in [Appendix B, Tables B8–B9](#). Instrument construction in the main specifications uses collapsed GMM style instruments with a t minus 2 to t minus 3 lag window for endogenous variables. [Appendix B, Table B1](#) reports the full diagnostic details.

- *p < 0.10, **p < 0.05, ***p < 0.01.

0.05, $p < 0.05$). Correlations among regressors are generally low to the highest is between DYN and CMX (0.22, $p < 0.01$), suggesting no immediate concerns regarding multicollinearity. Nevertheless, the multivariate specifications explicitly address endogeneity and state dependence. Descriptively, the observed dispersion (e.g., AIHC mean = 0.08; WAGE mean = 91.07 with substantial variance) underscores cross-firm heterogeneity, reinforcing the value of panel methods for inference.

5.2. Direct effect

[Table 3](#) reports the dynamic panel results for the direct, mediating, and moderating relationships among AI human capital (AIHC), wage inequality (WAGE), and firm value (Tobin Q). The lagged dependent variable is positive and highly significant in all specifications (e.g., 0.624 in Model 1; $t = 14.90$), confirming persistence in firm value and the appropriateness of the dynamic specification. Diagnostics are satisfactory across models (AR(2) $p > 0.18$; Hansen p between 0.35 and 0.43; instrument counts < groups), indicating no evidence of second-order

serial correlation and acceptable instrument validity. In Model 1, AIHC is positively associated with Tobin Q (0.065, $t = 2.50$), supporting [H1](#). The magnitude should be interpreted as an association: a one percentage point increase in the share of AI skilled workers is associated with an approximately 0.065 point higher Tobin Q. Model 6 shows that AIHC is negatively related to WAGE (−0.064, $t = 2.50$), supporting [H2](#). Model 2 indicates that WAGE is negatively associated with Tobin Q (−0.162, $t = 2.25$), supporting [H3](#).

5.3. Mediating effect

When WAGE is added to the Tobin Q equation (Model 2), the coefficient on AIHC remains positive but attenuates from 0.065 (Model 1) to 0.061 ($t = 1.84$), indicating that part of AI human capital's association with firm value operates through the wage-structure channel rather than solely through direct productivity and innovation effects. The product-of-coefficients estimate for the indirect path—AIHC → WAGE = −0.064 multiplied by WAGE → Tobin Q = −0.162—yields an indirect

effect of +0.010. Substantively, this implies that wage inequality decreases associated with higher AIHC account for a meaningful share of the overall AIHC value association. Using the difference-in-coefficients perspective, the direct effect declines by 0.004 once WAGE is controlled; the larger product-of-coefficients estimate is unsurprising in a system GMM context, where separate dynamic equations rely on different moment conditions and instrument sets. Taken together, these results indicate complementary partial mediation (H4): AIHC is associated with firm value both directly and indirectly through lower wage inequality, while the direct path remains positive after conditioning on WAGE (see Fig. 1).

5.4. Moderating effect

Moderation by industry environment (H5a–H5c). Model 3 indicates that the AIHC and Tobin Q association is stronger when industry dynamism (DYN) is higher. The AIHC $\times$ DYN interaction is positive and significant (0.012, $t = 2.40$), supporting H5a. Model 4 indicates that the payoff to AIHC is similarly larger when industry complexity (CMX) is higher (AIHC $\times$ CMX = 0.023, $t = 2.52$), supporting H5c. By contrast, Model 5 provides no evidence of an average linear munificence effect on the AIHC $\rightarrow$ value relationship (AIHC $\times$ MUN = 0.090, $t = 1.17$, n.s.), so the main-text version of H5b is not supported. Taken together, these patterns align with the view that AI-complementary capabilities are especially valuable when volatility and heterogeneity raise information-processing and coordination demands, whereas resource abundance alone is not systematically associated with larger returns. Fig. 2 visualizes the moderating roles of industry dynamism (DYN), complexity (CMX), and munificence (MUN) on the relationship between AI human capital (AIHC) and firm value (Tobin Q). Each plot shows predicted values across the range of AIHC for low ($\mu - 1\sigma$) and high ($\mu + 1\sigma$) levels of the moderator.

Because the null average munificence interaction remained theoretically interesting, Appendix B (Table B8) reports exploratory follow-up analyses that probe whether munificence operates in more conditional ways. Specifically, we introduce a curvilinear munificence term and joint interaction structures with environmental dynamism and complexity to assess whether resource-rich environments differentially shape the AI human capital (AIHC) to firm value relationship under

varying competitive conditions. These exploratory results suggest that munificence does not exert a uniform moderating effect but becomes more supportive of AIHC-driven value creation when combined with higher dynamism and complexity, consistent with the view that slack resources amplify AI returns when information-processing and reconfiguration demands are high. We interpret these appendix results as boundary-condition evidence rather than as a replacement for the main linear moderation test.

5.5. Additional analysis

Appendix B reports additional moderation analyses that extend the main framework by incorporating institutional, competitive, and firm-level boundary conditions. Specifically, we examine regulation intensity, international competition, labour cost intensity, and intangible asset intensity using the same dynamic system GMM specification as in the main analyses. The results show that regulatory intensity weakens, while competitive pressure, labour cost salience, and intangible asset intensity strengthen, the positive association between AI human capital and firm value. Marginal effects and a joint specification confirm the robustness of these patterns, indicating that the value creation association of AI human capital is contingent on both external constraints and firm-specific resource conditions.

5.6. Robustness check

To ensure that our main findings are not artifacts of model specification, measurement, or estimation choices, we conducted a series of robustness analyses. These tests varied the dependent variables, alternative measures of wage inequality, lag structures, sample composition, and estimators, added static firm fixed-effects benchmark models, and carefully monitored the risk of instrument proliferation. The full results are reported in Appendices C and D. Across these analyses, the key conclusions remain consistent: (i) AI human capital (AIHC) is positively associated with firm value, (ii) AIHC is negatively associated with wage inequality (WAGE), which in turn is negatively associated with firm value, providing evidence of partial mediation, and (iii) the AIHC value association is stronger in dynamic and complex industries but not in average linear munificent environments.

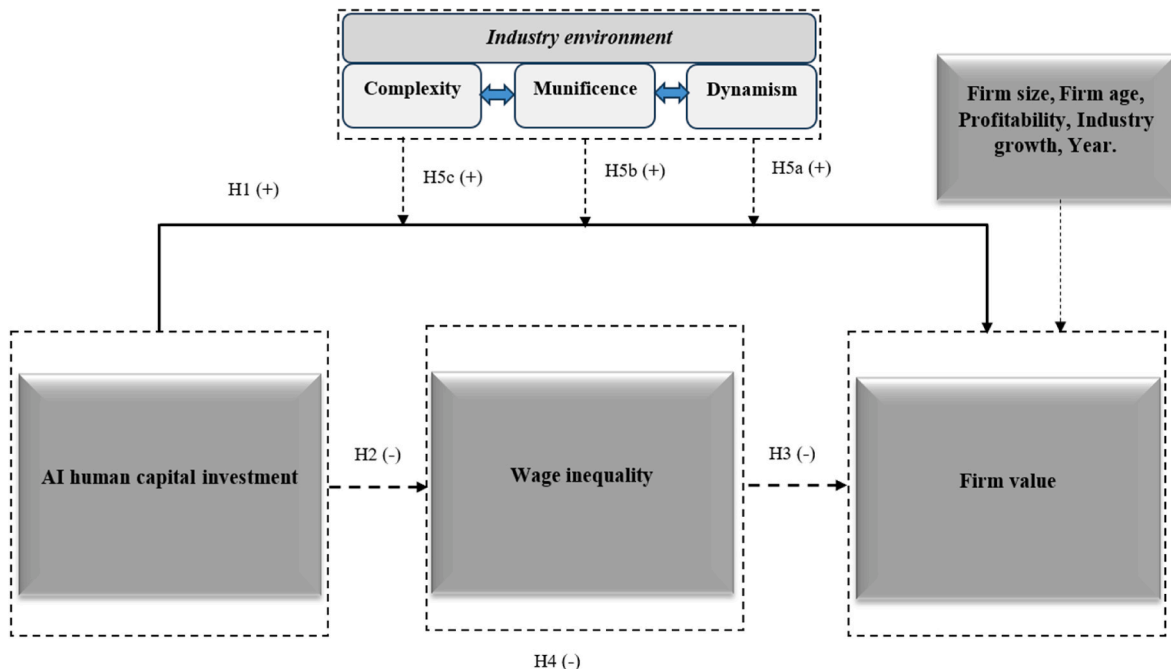


Fig. 1. Research model.

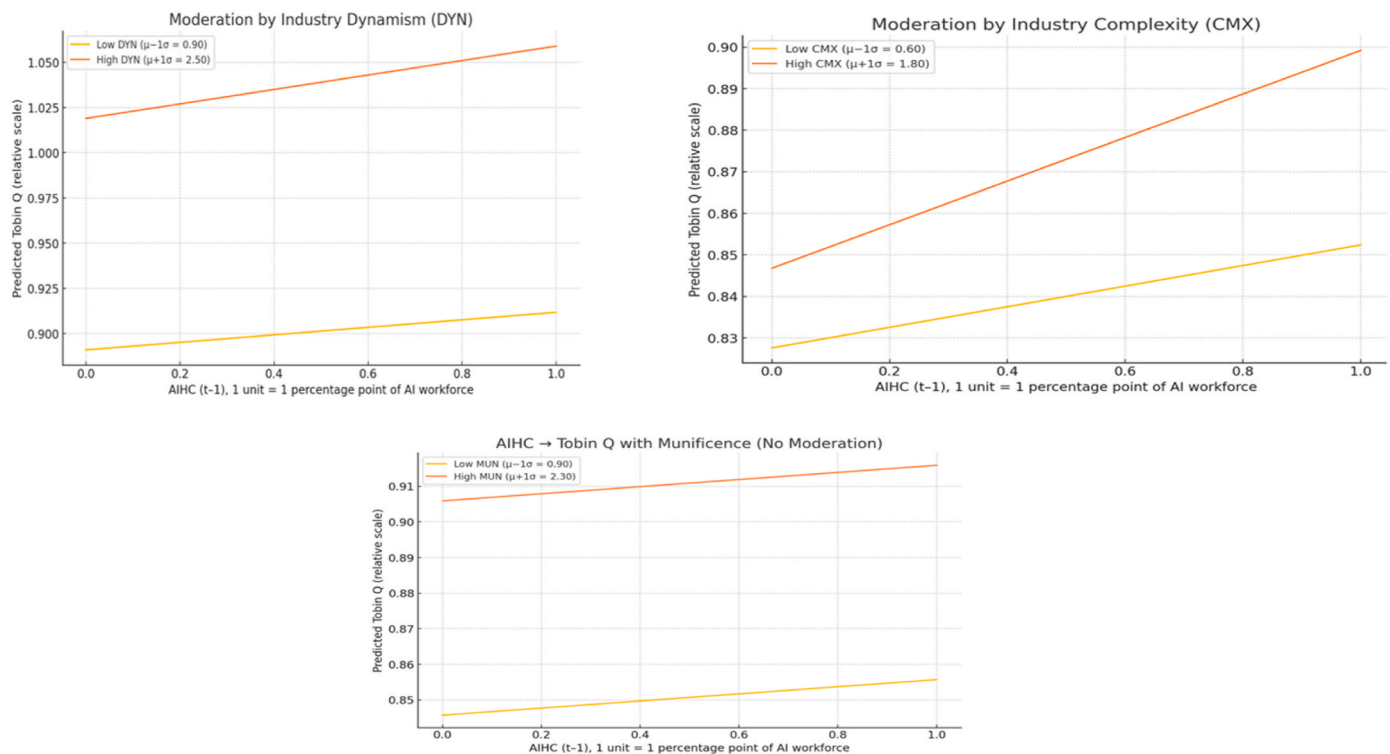


Fig. 2. Moderating effect.

Appendix C, Table C1 reports models that replace Tobin's Q with the Market-to-Book ratio (MB) as the outcome. The results confirm the robustness of our findings. AIHC remains positive and significant across specifications (0.012 in Model 1, 0.009 with WAGE included in Model 2), while WAGE is negatively associated with MB (-0.110 , $p < 0.05$). The $\text{AIHC} \times \text{DYN}$ interaction (0.010 , $p < 0.05$) and $\text{AIHC} \times \text{CMX}$ interaction (0.016 , $p < 0.05$) also remain positive and significant, reinforcing the moderating roles of dynamism and complexity. Similarly, Appendix C, Table C2 uses Return on Assets (ROA) as the outcome. Although the magnitudes are smaller as expected for an accounting-based measure, AIHC continues to predict higher ROA (0.003 , $p < 0.10$), while WAGE remains negative (-0.008 , $p < 0.05$). These results demonstrate that the findings are not limited to Tobin's Q and hold under both market-based and accounting-based measures of performance.

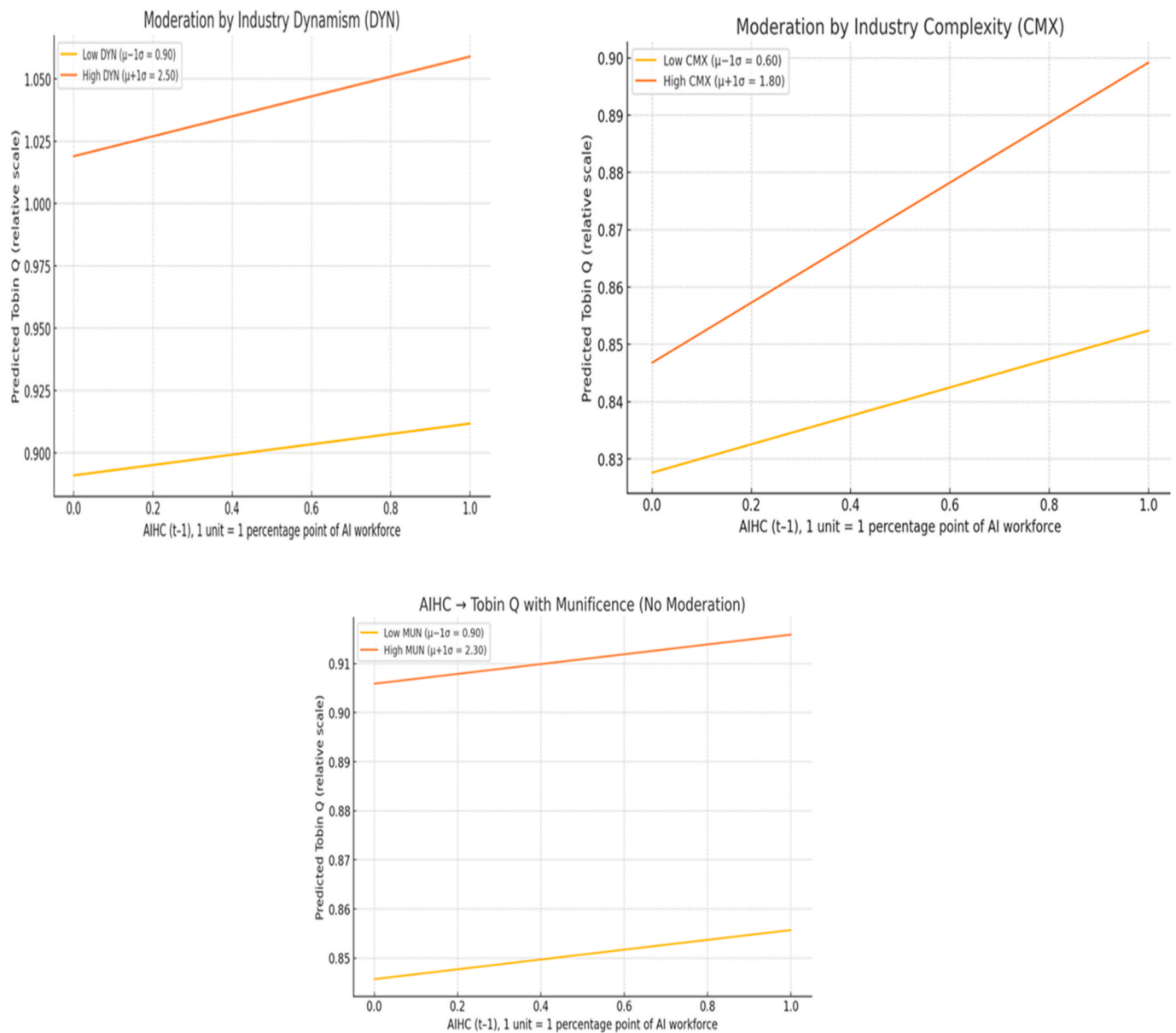
Appendix C, Table C3 explores whether the results are sensitive to the operationalization of wage inequality. Instead of using the Top Management Team (TMT) pay ratio, we substitute the CEO-to-employee pay ratio (WAGE_CEO). AIHC is negatively associated with WAGE_CEO (-0.025 , $p < 0.10$), and WAGE_CEO is negatively associated with Tobin Q (-0.095 , $p < 0.05$). While the magnitudes are smaller compared to the TMT measure, the pattern is consistent with our theoretical expectations and supports the robustness of the mediation pathway. This suggests that the mediation effect is not dependent on whether inequality is measured with a broader TMT reference group or focused solely on the CEO.

Appendix C, Table C4 tests the sensitivity of the results to timing conventions and sample composition. In the left panel, we lag AIHC by two years (t minus 2) to strengthen temporal interpretation by imposing a longer delay between AI investments and firm outcomes. AIHC remains positive (0.008 , $p < 0.10$), although the magnitude is somewhat reduced compared to the one-period lag. The mediation and interaction terms remain unchanged in sign and significance. In the right panel, we exclude the year 2020 to control for the disruptive effects of the COVID-19 pandemic on the tourism, travel, and hospitality industries. The AIHC coefficient (0.014 , $p < 0.05$) and the WAGE effect (-0.150 , $p < 0.05$)

remain consistent, with positive and significant interactions for DYN and CMX but no significant effect for MUN. This supports the view that our findings are not artifacts of pandemic specific shocks or the lag structure chosen.

Appendix C, Table C5 compares our preferred system GMM results with difference GMM (Arellano–Bond). The difference GMM results yield coefficients of similar sign and significance. AIHC is positive (0.013 , $p < 0.05$), and WAGE is negative (-0.148 , $p < 0.05$). Diagnostics (AR (2) and Hansen tests) remain within acceptable ranges. This reinforces that our findings are not dependent on the choice between system and difference GMM estimators, although the system GMM remains more efficient for our unbalanced panel. Finally, Appendix C, Table C6 addresses concerns about instrument proliferation, a common issue in GMM estimation. When uncollapsed instruments and the full lag structure are used, the number of instruments rises to 382 and the Hansen test p-value approaches 0.98, indicating probable overfitting and weakened validity of the test. In contrast, our preferred specification with collapsed instruments and capped lags yields 128 instruments and more credible Hansen p-values (0.41). A tighter lag cap further reduces instruments to 96 while maintaining consistent results. These tests support the robustness of our main findings to instrument design and that our reported results are based on the most reliable specification.

Appendix C, Table C12 additionally reports static firm fixed-effects benchmark models that reproduce the positive AIHC–value association, the negative AIHC–inequality association, and the negative inequality–value association, albeit with smaller magnitudes than the dynamic models. Across all robustness checks, the substantive conclusions remain intact. AIHC remains positively associated with firm value, WAGE remains negatively associated with value and is consistent with the mediation pathway, and the AIHC–value link is consistently amplified by dynamism and complexity but not by average linear munificence. The persistence of these results across outcomes, mediator definitions, lag structures, sample exclusions, estimators, and static panel benchmarks, together with instrument validity diagnostics, provides additional confidence in the reliability of our conclusions.



5.7. Sectoral heterogeneity: tourism vs. travel vs. hospitality

To examine whether the AI human capital association differs across subsectors, we developed sector-specific dynamic panel models and complemented them with pooled specifications that include sector interactions. [Appendix C, Table C7](#) reports the sector-stratified firm-value equations (DV = Tobin Q, controlling for WAGE), [Appendix C, Table C8](#) presents the sector-stratified mediator equations (DV = WAGE), [Appendix C, Table C9](#) pools all observations with sector interaction terms, [Appendix C, Table C10](#) summarizes marginal effects by sector, and [Appendix C, Table C11](#) provides Wald tests of coefficient equality across sectors.

[Appendix C, Table C7](#) reveals that AIHC is positively associated with Tobin Q in all three sectors, but the magnitude varies meaningfully. The coefficient is largest in Travel (0.018, $t = 2.45$), followed by Hospitality (0.012, $t = 1.87$) and Tourism (0.010, $t = 1.66$). In each sector, the lagged dependent variable remains large and significant ($=0.62$), confirming persistence in firm value. WAGE is consistently negative (-0.170 to -0.140 , $p < 0.05$), indicating that higher top-to-employee

pay dispersion is associated with a lower firm value within each sub-sector. Control variables show the expected patterns. PRFT is positive and significant ($=0.30$), whereas SIZE and Age are small and statistically weaker, while diagnostics (AR (2) $p = 0.24$ – 0.25 ; Hansen $p = 0.40$) indicate acceptable serial-correlation and instrument-validity properties across sectors.

Turning to the wage-structure channel, AIHC is significantly negatively associated with WAGE in all sectors, again with varying magnitudes (see [Appendix C, Table C8](#)). The inequality compression association is strongest in Hospitality (-0.060 , $t = 2.45$), followed by Travel (-0.042 , $t = 2.05$) and Tourism (-0.028 , $t = 1.72$). These patterns match the sector's operational profiles: hospitality is highly labor-intensive and knowledge-interdependent, so upgrading AI skills among non-executive personnel is plausibly accompanied by pay compression toward pivotal technical roles; travel and tourism exhibit the same direction but somewhat smaller magnitudes. Lagged WAGE is strongly persistent (0.70, $p < 0.01$), and diagnostics again fall in acceptable ranges.

The pooled models formally test sectoral heterogeneity. Using

Hospitality as the baseline, the AIHC $\times$ Travel interaction in the Tobin Q equation is positive and significant (Table 9), while the AIHC $\times$ Tourism interaction is small and statistically insignificant. The implied marginal effects are 0.018 (Travel), 0.012 (Hospitality), and 0.010 (Tourism). For the mediator equation, the baseline AIHC effect on WAGE is -0.060 in Hospitality; the positive interactions for Travel ($+0.018$) and Tourism ($+0.032$) indicate that AIHC remains associated with lower inequality in those sectors but to a lesser extent than in Hospitality. The corresponding marginal effects in Table 10 are -0.042 (Travel) and -0.028 (Tourism).

Wald tests confirm statistically meaningful cross-sector differences. For firm value, AIHC's association in Travel exceeds Hospitality ($\chi^2 = 4.21$, $p = 0.040$) and Tourism ($\chi^2 = 5.41$, $p = 0.020$), whereas Tourism does not differ significantly from Hospitality ($p = 0.283$) (Table 11). For wage inequality, the reduction associated with AIHC is strongest in Hospitality; it is significantly larger than in Tourism ($\chi^2 = 4.73$, $p = 0.030$) and marginally larger than in Travel ($\chi^2 = 3.06$, $p = 0.080$). Taken together, these tests align with the sector-stratified and pooled values.

The sectoral pattern is consistent with our broader theoretical account. In Travel, elevated market volatility and coordination demands likely make AI-enabled capabilities especially valuable, yielding the largest AIHC–Tobin Q association. In Hospitality, the stronger inequality-compression effect of AIHC is in line with the sector's labour-intensive service model, where complementing frontline work with AI skills improves internal equity and collaboration. Tourism exhibits positive but more moderate effects on both outcomes. Importantly, these differences do not overturn the study's core conclusions: across all subsectors, AIHC remains positively related to firm value, negatively related to wage inequality, and the adverse association between inequality and firm value persists. Diagnostics (AR and Hansen) remain within acceptable bounds, and instrument counts are kept well below the number of firms, supporting the reliability of the sector-specific values.

Appendix D reports robustness checks using alternative measures of AI human capital that capture both hiring-based investment and a stricter stock of core AI talent. Using AI job posting share, AI hiring intensity, and a conservative core AI role share, the dynamic system GMM results remain consistent with the main findings: AI human capital is positively associated with firm value and negatively related to wage inequality, with stronger effects in more dynamic and complex environments. Prevalence and overlap analyses further show that these alternative measures capture meaningful variation over time and across segments, reinforcing the interpretation that the results reflect AI capability building rather than measurement artifacts tied to a specific proxy.

6. Discussion and implications

6.1. Key findings

Our dynamic panel analysis yields four principal findings. First, AI human capital intensity exhibits a positive association with firm value (Tobin's Q), net of persistence and controls. This result is consistent with the resource-based view. AIHC is a scarce, firm-embedded capability that, when combined with data assets and complementary routines, generates rents observable in market valuations (Barney, 1991; Peteraf, 1993). It also accords with recent evidence that AI intensity is associated with growth and innovation (Babina et al., 2024) and with service scholarship on AI-enabled productivity and customer experience (Huang & Rust, 2018). Within tourism, travel, and hospitality, prior work documents value creation through analytics-driven personalization and operational automation (Buhalis & Law, 2008; Gretzel et al., 2015; Ivanov & Webster, 2019a, 2019b).

Second, AIHC is associated with lower wage inequality (WAGE), and WAGE, in turn, is negatively related to Tobin's Q, yielding a statistically

significant partial mediation. The WAGE $\rightarrow$ value link aligns with research showing that wide pay dispersion undermines collaboration, quality, and retention (Bloom, 1999; Cowherd & Levine, 1992; Pfeffer & Langton, 1993). At the same time, our evidence contrasts with a strict tournament-theory expectation that higher vertical dispersion should enhance incentives and performance (Bamberger et al., 2021; Connelly et al., 2014). We contend that the service-system interdependence that typifies TTH moderates these incentive benefits: where task coupling and knowledge sharing are high, dispersion harms team production (Shaw et al., 2002), making equity effects salient. AIHC may compress inequality by raising the bargaining power and contribution of technical and mid-tier employees and by pushing firms to redesign workflows around human–AI collaboration mechanisms consistent with complementarity arguments in the technology–human capital literature (Acemoglu & Restrepo, 2019).

Third, the AIHC–value relationship strengthens with environmental dynamism and complexity, while the average linear munificence interaction is not supported. These interactions cohere with contingency and dynamic-capabilities logics: under turbulence and heterogeneity, analytics, experimentation, and cross-functional orchestration expand AIHC's marginal return (Dess & Beard, 1984; Keats & Hitt, 1988; Teece, 2007). By contrast, the null average munificence effect mirrors arguments that slack can relax selection pressures and allow slower diffusion of best practices (Castrogiovanni, 1991). Our exploratory appendix analyses suggest that munificence becomes more supportive only when it coexists with high dynamism or complexity, which helps reconcile why slack does not systematically translate into stronger AI returns on its own. Finally, sectoral analyses reveal meaningful heterogeneity. AIHC's association with Tobin's Q is largest in Travel, which faces volatile demand and multi-sided distribution, and remains positive in Hospitality and Tourism; by contrast, AIHC's inequality-reducing effect is strongest in Hospitality, a labour-intensive, service-rich context. These patterns are compatible with theory in which travel firms exploit AIHC for agile pricing, routing, and capacity decisions amid high uncertainty, whereas hospitality firms realize equity and coordination gains from diffusing AI skills across interdependent frontline roles. Across sectors, however, the core mechanisms are invariant. AIHC is associated with higher value both directly and indirectly through lower inequality, and environmental turbulence is associated with larger payoffs.

6.2. Theoretical implications

Our findings advance theory at the intersection of the resource-based view (RBV), pay dispersion and tournament logics, and contingency/dynamic capabilities in service industries. First, we reconceptualize AI human capital, the firm-embedded stock of employees with AI-related skills as an RBV microfoundation. AIHC is valuable because it enables data-driven problem solving, experimentation, and human–AI orchestration. It is rare and imperfectly tradable because skills are co-specialized with firm data, workflows, and domain knowledge and it is difficult to imitate because benefits hinge on complementarities with routines and governance (Barney, 1991; Clark et al., 2026; Peteraf, 1993; Sirmon et al., 2007). The positive AIHC–value association we document extends RBV by specifying the human-capital pathway through which AI capability is associated with market valuation in TTH, complementing evidence that AI intensity fuels growth and innovation (Babina et al., 2024) and service research on AI-enabled productivity and experience (Huang & Rust, 2018; Ivanov & Webster, 2019a, 2019b; Marchesani & Testa, 2026; Mostafiz et al., 2025; Wang et al., 2025).

Second, by showing that wage inequality (WAGE) partially mediates the AIHC–value link, we integrate RBV with socio-structural accounts of performance in services. Prior work documents that vertical pay dispersion can erode collaboration, knowledge sharing, and quality (Bloom, 1999; Campos-Soria et al., 2009; Casado-Díaz & Simón, 2016; Cowherd & Levine, 1992; Pfeffer & Langton, 1993), counter to strict tournament predictions of stronger incentives (Connelly et al., 2014; de

Castro Romero, Barroso, & Santero-Sánchez, 2025; Marfil-Cotilla et al., 2024). Our evidence identifies boundary conditions for tournament theory, where task interdependence and co-production are high, dispersion's incentive benefits are dominated by coordination and fairness costs (Shaw et al., 2002). AIHC may compress WAGE by elevating the contribution and bargaining power of technical and mid-tier roles and by prompting work redesign around human-AI complementarities (Acemoglu & Restrepo, 2019; Guimarães & Silva, 2016; Lee & Kang, 1998), improving performance indirectly and directly.

Third, the moderating effects refine contingency and dynamic-capabilities arguments. We find that environmental dynamism and complexity amplify the payoff to AIHC, consistent with the view that sensing, seizing, and reconfiguring depend on information processing and cross-functional integration when uncertainty and heterogeneity are high (Dess & Beard, 1984; Keats & Hitt, 1988; Teece, 2007). By contrast, munificence does not reliably strengthen the AIHC-value link in the average linear model, aligning with the idea that slack may attenuate selection pressures and slow diffusion of process innovations (Castrogiovanni, 1991). The exploratory appendix evidence nevertheless suggests a more conditional role for munificence when resource richness is paired with higher dynamism or complexity. These results specify when AIHC is most consequential and caution against assuming resource abundance alone magnifies returns. Fourth, sectoral heterogeneity offers theoretically meaningful boundary conditions for generalization. AIHC's association with firm value is strongest in Travel, where volatility, networked distribution, and capacity coordination heighten the value of predictive and integrative skills, and its inequality-reducing effect is greatest in Hospitality, a labour-intensive context where team production is central. The pattern suggests that service architecture, customer contact, modularity, and interdependence condition whether AIHC creates value mainly through market agility or internal equity and coordination.

Finally, our study contributes methodologically to theorizing capabilities by operationalizing AIHC with a labor-based measure and embedding mediation and moderation within a dynamic panel framework. Using lag structure and system GMM to address persistence and endogeneity links microfoundations to outcomes without external instruments (Roodman, 2009). Conceptually, positioning wage inequality as a mechanism clarifies how socio-technical transformation translates to value, connecting RBV, pay dispersion, and contingency perspectives in services.

6.3. Managerial implications

Building on our dynamic panel results, several actionable implications emerge for senior managers in tourism, travel, and hospitality. First, the positive association between AI human capital and firm value suggests that firms should treat AI-related skills as a strategic asset rather than a discrete IT purchase. This extends beyond hiring a few data scientists. Hotels and airlines benefit from hybrid roles that combine domain expertise with analytics (e.g., revenue managers who code and operations leaders who interpret models). Hotel groups deploy demand-forecasting and price-optimization engines to align rates with booking curves, while airlines use machine-learning systems to adjust fares and ancillary bundles, both rely on staff who translate model outputs into decisions. Online travel intermediaries likewise personalize search through recommenders and experimentation. Success requires a shared analytics vocabulary across product, engineering, and data teams.

Second, because wage inequality (WAGE) is negatively associated with firm value and partially mediates the AIHC-value link, leaders should anticipate and manage distributional consequences as AI diffuses. Redesign career ladders and pay bands so that AI-complementary frontline and mid-tier roles gain recognition and progression. Productivity gains from AI-assisted scheduling and service-recovery that are more durable when paired with upskilling budgets, certification pathways, and transparent bonuses tied to team outcomes rather than

individual sales. Tracking an equity dashboard pay ratios, promotion velocity, internal mobility can alert executives to unintended widening of dispersion during transformation.

Third, our moderation results imply that context matters. Where demand and product complexity are high, hub-and-spoke airline networks, multi-property hotel portfolios, multi-channel tour operators the returns to AIHC are amplified. Managers in such environments should prioritize use cases that exploit uncertainty (e.g., near-real-time pricing, dynamic capacity allocation, and predictive staffing that flexes house-keeping, concierge, and check-in resources by arrival pattern). By contrast, munificent environments do not automatically strengthen AI payoffs, thus, slack can mask execution problems and slow learning. Governance should therefore emphasize disciplined test-and-learn cycles, post-mortems, and clear stage-gates for scaling pilots.

Fourth, sectoral heterogeneity points to distinct roadmaps. In Travel, where volatility and network coordination are first-order constraints, emphasize AIHC for integrated revenue management, disruption recovery, and irregular-operations decision support. In Hospitality, prioritize broad workforce upskilling AI-assisted housekeeping routing, personalized service scripts, and demand-based labor planning paired with equity safeguards that keep teams cohesive. In Tourism attractions, focus on guest-flow forecasting, dynamic ticketing windows, and AI-aided crowd management to protect experience quality without over-staffing.

Fifth, societally, fairer pay structures and predictable operations enhance worker wellbeing and reduce waste and emissions through better forecasting and capacity use. By providing evidence that AI human capital is associated with lower wage inequality and higher firm value, our study motivates investments that spread AI skills beyond a small elite, improving dignity at work, career progression, and retention. More accurate demand prediction allows hotels and airlines to schedule staffing, energy, and inventory to actual needs, cutting food waste, idle energy loads, and unnecessary trips. These changes strengthen household income stability, service reliability, and local destination resilience, while advancing inclusive, low-carbon growth.

Finally, sequence implementation: (1) build a minimally sufficient data foundation (clean reservations, POS, and operations data; clear ownership); (2) appoint a cross-functional AI steering group that includes HR/compensation; (3) start with two or three high-ROI use cases with pre-registered evaluation plans and success metrics (RevPAR, ancillary revenue per passenger, guest-satisfaction recovery); (4) institutionalize capability transfer communities of practice, code repositories, and bite-size training to avoid dependency on a few experts. These steps translate the study's message into practice: invest in AIHC, redesign work and rewards to sustain equity, tailor to environmental complexity and dynamism, and scale through disciplined learning.

6.4. Implications for policymakers

Findings from this study yield several implications for policymakers seeking to enhance productivity, equity, and resilience in tourism, travel, and hospitality. First, because AI human capital is positively associated with firm value, skills policy should prioritize human-capital formation alongside digital capital. Targeted training tax credits, apprenticeships in data engineering and revenue analytics, and industry university programmes can accelerate AIHC accumulation, especially for SMEs facing fixed-cost barriers. Real-world deployments, hotel demand forecasting, airline fare optimization and disruption recovery, and timed-entry management at attractions require hybrid roles blending domain knowledge and analytics and public policy can underwrite these pipelines at scale.

Second, the evidence that wage inequality (WAGE) is negatively associated with firm value, and partially mediates the AIHC-value link, suggests that inclusive AI diffusion may be economically beneficial. Policymakers should attach training and pay progression conditionalities to AI adoption grants (e.g., certification pathways for frontline and

mid-tier roles using AI tools, team-based bonuses). Requiring firms to track equity dashboards, pay ratios, promotion velocity, internal mobility can deter widening dispersion. Given pronounced inequality compression in hospitality, hospitality-specific upskilling vouchers and recognition of AI-complementary competencies are likely to yield outsized returns.

Third, moderation results indicate that dynamism and complexity are associated with larger AIHC returns, whereas munificence does not. Consequently, generic subsidies are less effective than context-sensitive instruments. Policymakers should prioritize high-uncertainty use cases dynamic pricing, predictive staffing, capacity allocation through competitive challenge funds and regulatory sandboxes. In parallel, investment in data collaboratives linking bookings and POS data would enable smaller operators to participate in AI benefits otherwise limited to large platforms.

Fourth, algorithmic governance must keep pace with commercial practice. Consumer-facing safeguards algorithmic impact assessments for personalized pricing and external audits for algorithmic collusion can preserve trust while allowing innovation. Agencies should clarify acceptable data-sharing and pricing conduct for OTAs, airlines, and hotel consortia to reduce legal uncertainty.

Finally, policy should respect sectoral heterogeneity. In travel, investment in operations-research capability for disruption management has high social returns. In hospitality, where our evidence indicates stronger inequality compression, emphasis should fall on broad workforce upskilling and progression. Tourism attractions benefit from grants for guest-flow analytics and timed-entry systems that protect experience quality. Together, these instruments translate the evidence into a policy portfolio that scales AI value, safeguards equity, and fosters competitive markets.

7. Limitations and future research directions

Several limitations suggest caution in interpretation and point to fruitful avenues for future inquiry. First, a key limitation of this study is that the evidence is based on UK-listed tourism, travel, and hospitality firms, and the magnitude and form of the observed relationships may depend on institutional features of the UK setting. The UK context combines comparatively strong disclosure and governance requirements for listed firms and market-based valuation mechanisms, which may heighten the visibility of AI human capital effects in firm value and improve the measurement of executive pay structures, while labour-market institutions and wage-setting practices may shape how AI-related skills translate into within-firm pay dispersion. Accordingly, although the underlying mechanisms linking AI human capital intensity, wage inequality, and firm value are theoretically applicable to service economies more broadly, their empirical expression may differ in countries with more coordinated wage bargaining, stronger employment protection, or different regulatory and capital-market regimes. Future research should therefore replicate and extend our design across institutional contexts and sectors to test boundary conditions and assess whether AI human capital generates similar performance and distributional implications beyond the UK and beyond service-intensive environments. Second, although we model dynamics using internal-instrument system GMM and benchmark the core relationships against static firm fixed-effects panels, observational data cannot fully resolve remaining concerns about reverse causality, measurement error, and instrument weakness. We do not claim a separately identified external-IV design for AIHC, because any exclusion restriction in this setting would be difficult to defend. Dynamic panel estimators are sensitive to instrument proliferation, stationarity assumptions, and lag-selection choices. While our diagnostics are acceptable, small departures from these assumptions can bias estimates (Roodman, 2009). Future research could complement our approach with designs that exploit quasi-experimental variation (e.g., staggered regulatory shocks, cloud-cost shocks, or instruments based on exogenous exposure to AI

toolsets), as well as with matched difference-in-differences at the sub-sector level. Consistent with this limitation, the manuscript now interprets the estimated coefficients as robust conditional associations rather than definitive causal effects. This wording is deliberate because the system GMM design improves temporal ordering and addresses several endogeneity concerns, but it cannot fully substitute for exogenous assignment of AI human capital.

Third, measurement constraints warrant further work. Our AI human capital indicator captures the share of AI-skilled workers but not the depth of their expertise, the quality of human-AI orchestration, or complementarities with data assets and routines. Likewise, wage inequality (WAGE) is constructed from financial disclosures and may miss within-grade dispersion and non-wage forms of reward. Richer micro-data linking payroll, task assignments, and skill taxonomies would illuminate how resources are orchestrated into capabilities and outcomes (Sirmon et al., 2007). Future studies could disentangle distinct AIHC bundles (e.g., data engineering vs. ML operations vs. human-AI interface design) and test how specific complements (data governance, experimentation cadence, frontline autonomy) condition performance and equity. Fourth, external validity is bounded. Our context is UK-listed tourism, travel, and hospitality firms. Governance and remuneration practices, labour institutions, and platform competition differ across countries and among small and medium-sized enterprises. Comparative, multi-country panels would enable tests of institutional contingencies and cultural moderators. In addition, the 2018–2024 window spans pandemic and recovery phases that may alter the returns to AIHC and the salience of wage structures. Replication in more stable periods and in adjacent service domains (e.g., retailing, logistics) would help separate structural effects from shock-specific dynamics. Finally, our moderation tests use standard proxies for dynamism, munificence, and complexity; these may be coarse relative to the fine-grained environmental turbulence faced by multi-channel service firms. Future work could integrate digital-market metrics (e.g., search volatility, platform algorithm updates) and customer-journey heterogeneity, and model how dynamic capabilities sensing, seizing, reconfiguring mediate the translation of AIHC into value under varying conditions (Teece, 2007). Longitudinal case–panel hybrids, combining archival data with process tracing inside firms, would clarify the mechanisms linking AIHC, wage structures, and firm value over time.

CRediT authorship contribution statement

Gomaa Agag: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Amna Al Zaabi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Riyad Eid:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Impact statement

This study explains how tourism, travel, and hospitality firms can pursue artificial intelligence without widening organizational inequality. It shows that stronger AI human capital is associated with higher market value and lower internal wage inequality, especially where service environments are dynamic or complex. The findings support investment in AI skills beyond a small technical elite by linking workforce capability, fairer pay structures, service reliability, and competitiveness. For managers, the results encourage job redesign, AI upskilling, and transparent progression pathways. For policymakers, they support training incentives and inclusive AI diffusion in labour intensive service sectors. For destinations and communities, broader AI skills can improve service quality, operational resilience, productivity,

and stable employment. The paper therefore contributes to the real world challenge of combining AI led competitiveness with fairer work and more resilient tourism economies.

Declaration of competing interest

The author declares that there is no conflict of interest.

Acknowledgements

(None)

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tourman.2026.105470>.

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